

**BOWIE, TX, USA**WMO: **720622**Lat: **33.602N**Lon: **97.776W**Elev: **1101**StdP: **14.12**Time Zone: **-6.00 (NAC)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.0       | 24.1 | 7.0                           | 8.2 | 25.4 | 11.8 | 10.4 | 34.6 | 24.8                  | 44.2 | 22.8 | 46.8 | 6.0                   | 0    | 0.444 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 22.3                   | 100.3           | 71.2 | 98.7 | 71.8 | 95.5 | 73.2 | 78.2                | 89.7 | 77.2 | 89.3 | 76.4 | 88.3 | 8.4                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.2                            | 137.6 | 82.9 | 73.9 | 131.5 | 81.9 | 73.0 | 127.6 | 81.4 | 42.4          | 89.4 | 41.4 | 89.4 | 40.5 | 88.5 | 81.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |
| 21.2              | 18.8 | 17.0 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A  |      |
|                   |      |      | DB                         |      |                    |     |                                                    |     |            |     |            |     |            |      |      |
|                   |      |      | WB                         | 11.3 | 79.6               | 5.2 | 1.8                                                | 7.5 | 80.9       | 4.4 | 82.0       | 1.5 | 83.0       | -2.3 | 84.3 |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 64.1   | 43.2  | 47.5  | 56.5  | 63.5  | 70.4  | 79.8  | 82.7  | 83.4  | 76.3  | 63.8  | 55.3  | 45.6 |
|                                                                                  | DBStd       | 16.23  | 9.72  | 11.64 | 9.74  | 8.01  | 6.98  | 4.74  | 6.64  | 4.83  | 6.30  | 9.13  | 8.75  | 9.41 |
|                                                                                  | HDD50       | 741    | 253   | 171   | 47    | 8     | 0     | 0     | 2     | 0     | 0     | 13    | 47    | 200  |
|                                                                                  | HDD65       | 2676   | 677   | 499   | 293   | 118   | 31    | 0     | 12    | 0     | 5     | 131   | 307   | 603  |
|                                                                                  | CDD50       | 5881   | 42    | 99    | 249   | 413   | 634   | 893   | 1016  | 1036  | 789   | 440   | 205   | 65   |
|                                                                                  | CDD65       | 2343   | 1     | 8     | 30    | 73    | 200   | 443   | 561   | 571   | 344   | 94    | 15    | 3    |
|                                                                                  | CDH74       | 26536  | 14    | 78    | 268   | 683   | 1901  | 4969  | 7015  | 7023  | 3439  | 978   | 144   | 24   |
|                                                                                  | CDH80       | 13140  | 0     | 19    | 56    | 169   | 659   | 2435  | 3903  | 3946  | 1612  | 308   | 31    | 2    |
| Wind                                                                             | WSAvg       | 7.0    | 7.2   | 8.0   | 8.8   | 8.9   | 7.6   | 6.7   | 5.5   | 5.2   | 5.3   | 6.6   | 7.3   | 7.1  |
| Precipitation                                                                    | PrecAvg     | 34.4   | 1.6   | 2.1   | 2.8   | 3.1   | 4.2   | 3.8   | 2.5   | 2.4   | 3.1   | 3.1   | 2.1   | 2.0  |
|                                                                                  | PrecMax     | 46.1   | 3.6   | 6.7   | 7.6   | 8.7   | 8.9   | 10.6  | 6.4   | 5.9   | 6.7   | 7.6   | 7.2   | 7.2  |
|                                                                                  | PrecMin     | 21.7   | 0.0   | 0.0   | 0.0   | 0.4   | 0.6   | 1.1   | 0.2   | 0.0   | 0.2   | 0.0   | 0.2   | 0.0  |
|                                                                                  | PrecStd     | 7.0    | 1.0   | 1.9   | 1.9   | 2.0   | 2.1   | 2.4   | 1.9   | 1.7   | 1.9   | 2.2   | 2.0   | 1.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 76.7  | 82.1  | 85.6  | 89.8  | 93.2  | 99.4  | 105.1 | 105.9 | 99.4  | 92.0  | 84.1  | 77.3 |
|                                                                                  |             | MCWB   | 55.2  | 63.3  | 63.9  | 66.1  | 73.7  | 70.9  | 70.4  | 69.9  | 68.2  | 69.3  | 65.4  | 60.3 |
|                                                                                  | 2%          | DB     | 70.2  | 75.2  | 81.1  | 84.3  | 90.0  | 96.8  | 100.1 | 101.9 | 94.9  | 86.8  | 77.6  | 70.7 |
|                                                                                  |             | MCWB   | 54.0  | 59.1  | 62.0  | 65.9  | 71.4  | 72.0  | 71.0  | 70.8  | 72.0  | 68.6  | 61.6  | 57.2 |
|                                                                                  | 5%          | DB     | 64.3  | 70.7  | 76.8  | 81.2  | 86.6  | 94.0  | 98.4  | 99.0  | 91.5  | 83.1  | 73.4  | 66.1 |
|                                                                                  |             | MCWB   | 52.2  | 58.2  | 61.1  | 65.1  | 69.9  | 73.5  | 72.0  | 72.5  | 72.2  | 66.5  | 60.8  | 55.5 |
|                                                                                  | 10%         | DB     | 59.7  | 65.9  | 72.4  | 78.0  | 83.8  | 91.1  | 95.1  | 96.0  | 89.3  | 79.6  | 70.2  | 61.4 |
|                                                                                  |             | MCWB   | 50.0  | 56.4  | 59.0  | 63.7  | 69.9  | 73.8  | 74.0  | 73.5  | 71.1  | 64.8  | 59.2  | 53.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 62.4  | 66.4  | 68.4  | 71.4  | 76.9  | 79.7  | 79.5  | 78.7  | 77.5  | 74.1  | 70.8  | 65.0 |
|                                                                                  |             | MCDB   | 68.0  | 73.6  | 78.9  | 80.1  | 86.4  | 90.0  | 90.5  | 89.1  | 89.1  | 83.9  | 77.1  | 71.2 |
|                                                                                  | 2%          | WB     | 57.3  | 64.1  | 66.2  | 69.8  | 74.6  | 77.8  | 78.2  | 77.4  | 76.2  | 72.7  | 66.2  | 61.3 |
|                                                                                  |             | MCDB   | 63.8  | 71.0  | 75.8  | 79.3  | 84.0  | 88.6  | 90.0  | 90.6  | 87.1  | 81.9  | 71.6  | 67.1 |
|                                                                                  | 5%          | WB     | 53.9  | 61.5  | 64.1  | 68.2  | 73.1  | 76.6  | 77.3  | 76.5  | 75.0  | 70.8  | 64.1  | 58.3 |
|                                                                                  |             | MCDB   | 63.6  | 67.7  | 71.5  | 77.3  | 82.1  | 87.6  | 89.5  | 89.8  | 85.4  | 79.1  | 69.2  | 63.1 |
|                                                                                  | 10%         | WB     | 50.4  | 58.0  | 62.3  | 66.6  | 71.5  | 75.4  | 76.4  | 75.5  | 73.9  | 68.4  | 61.7  | 53.7 |
|                                                                                  |             | MCDB   | 58.9  | 64.9  | 69.4  | 74.7  | 80.2  | 86.3  | 88.5  | 88.8  | 83.8  | 75.7  | 68.8  | 59.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 22.1  | 21.8  | 22.6  | 22.9  | 20.2  | 20.5  | 22.2  | 22.3  | 21.1  | 23.0  | 23.1  | 19.9 |
|                                                                                  |             | MCDBR  | 30.3  | 27.9  | 26.4  | 26.5  | 24.4  | 23.6  | 26.0  | 25.8  | 24.5  | 26.6  | 26.3  | 27.2 |
|                                                                                  |             | MCWBR  | 17.0  | 14.4  | 12.1  | 12.6  | 10.2  | 6.6   | 6.2   | 6.3   | 7.6   | 11.5  | 13.3  | 15.3 |
|                                                                                  | 5% WB       | MCDBR  | 25.9  | 23.8  | 21.4  | 22.0  | 19.4  | 19.9  | 20.5  | 21.3  | 19.1  | 21.1  | 19.9  | 21.1 |
|                                                                                  |             | MCWBR  | 17.4  | 15.7  | 12.1  | 12.6  | 9.6   | 7.9   | 6.9   | 6.7   | 6.7   | 11.1  | 14.5  | 15.0 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.299  | 0.309 | 0.343 | 0.374 | 0.413 | 0.427 | 0.427 | 0.421 | 0.386 | 0.349 | 0.320 | 0.303 |      |
|                                                                                  | taud        | 2.514  | 2.495 | 2.419 | 2.358 | 2.313 | 2.311 | 2.347 | 2.334 | 2.390 | 2.477 | 2.496 | 2.499 |      |
|                                                                                  | Ebn at Noon | 292    | 300   | 296   | 290   | 278   | 272   | 272   | 272   | 278   | 282   | 281   | 283   |      |
|                                                                                  | Edh at Noon | 26     | 30    | 35    | 39    | 41    | 41    | 40    | 40    | 36    | 30    | 27    | 25    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 927    | 1109  | 1458  | 1773  | 1907  | 2134  | 2155  | 1977  | 1643  | 1280  | 984   | 804   |      |
|                                                                                  | RadStd      | 95     | 170   | 100   | 146   | 198   | 200   | 142   | 115   | 151   | 189   | 127   | 95    |      |

**Historical Trends**

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